Firmware Signature Guide

This document guides customers on using the tool to generate RSA2048 keys, inject public keys, and create signature firmware for TMS upgrades.

1. Tool list

Number	Tool name	Purpose
1	RSA2048KeyGen	Generate an RSA2048 public and private key parameter file
2	WritePublicKey	Write the public key to the device
3	SignPackTool	Sign the firmware.img file to generate a .sig file

Note: During the debugging phase, the public key can be manually injected; for production devices, the production line tool will automatically inject the public key during manufacturing.

2. Overview of the process

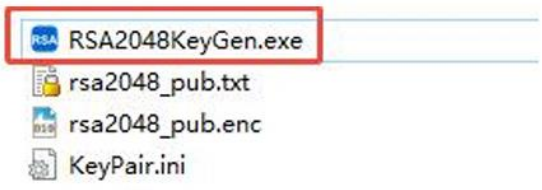
Generate public and private keys → Inject the public key → Sign the firmware

3. Detailed instructions

3.1. Generate RSA2048 public and private keys

3.1.1 Operating Tools

1. Navigate to the tool directory and find RSA2048KeyGen.exe;
2. Double-click to run;



3. The tool generates three files in the current directory by default:

KeyPair.ini
rsa2048_pub.txt
rsa2048_pub.enc

3.1.2 Output File Description

File Name	Content	Purpose
KeyPair.ini	Private key parameters, including: N、E、P、Q、DP、DQ、Qinv、D	Used for firmware signing, must be kept strictly confidential
rsa2048_pub.txt	Public key in plaintext, containing N and E	Used for injecting during the production phase
rsa2048_pub.enc	Public key ciphertext, containing N and E	Used for injecting devices during the debugging phase

3.1.3 Precautions

1.KeyPair.ini is the private key parameter file and must not be disclosed under any circumstances. If it gets lost, the firmware cannot be re-signed.

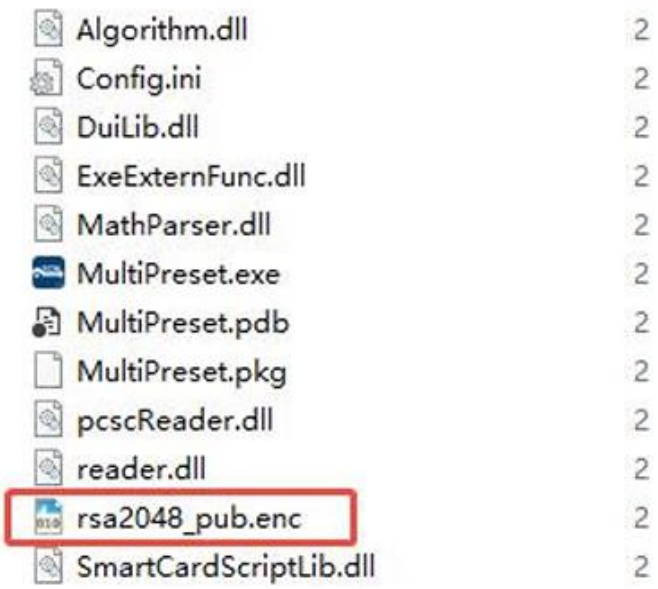
2. rsa2048_pub.txt can be made public and used to inject devices during production.

3. It is recommended to back up both files to a secure directory and record the creation date and purpose

3.2 Injecting the public key

3.2.1 Debugging Phase

1. Copy the rsa2048_pub.enc file generated in step 3.1 to the 2.WritePublicKey directory. Make sure not to use the default rsa2048_pub.enc file in the tool directory;



2. Navigate to the 2.WritePublicKey directory and run MultiPreset.exe;
3. After successful writing, the device completes the public key writing process.

3.2.2 Production Phase

During production, the official device automatically has its public key injected by the production line tool, so customers don't need to do anything manually.

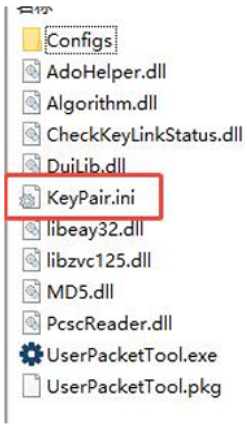
3.2.3 Precautions

1. The public key only needs to be injected once; re-injection may prompt "Public Key Exists".
2. If it prompts "Public key validation failed", please confirm whether the device is properly connected.
3. The public key cannot be changed after injection, unless the device is erased or a dedicated recovery process is used.
4. Manual injection is allowed during the debugging phase; please use the production line tool during production to avoid errors.

3.3 Signed Firmware

3.3.1 Prepare the files

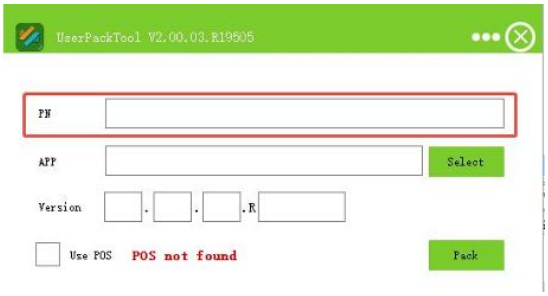
1. Copy the KeyPair.ini generated in step 1 to 3.TMS_SignPackTool;



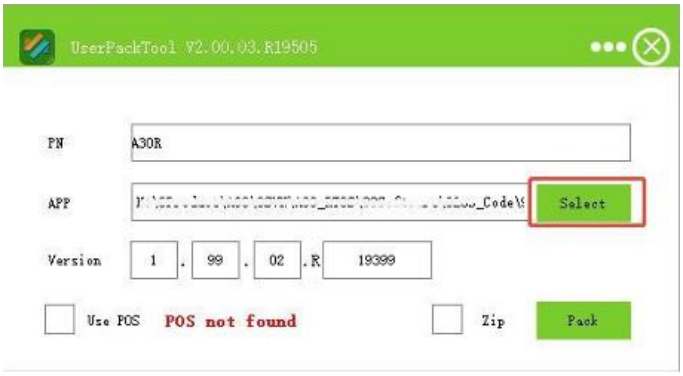
2. Prepare the firmware img file to be signed, for example: A30R_APP_V1.99.02.R19399.img.

3.3.2 Steps to Perform

1. Open UserPacketTool.exe:
2. Select the corresponding product model:



3. Select the image file to be signed. If the image follows the correct naming convention, the tool will automatically parse the version number:



4. Check the "Zip" box, and the tool will use the compressed version of the img file for packaging. Click Pack to generate the firmware package required for TMS upgrade.



5. The tool automatically reads the KeyPair.ini file and signs the img;
6. After signing is complete, a .sig signature file is generated in the same directory, for example: pkg_A30R_V01.99.02.R19399_0_0_TMS.sig.

3.3.3 Input and Output File Descriptions

File Name	Explanation
A30R_APP_V1.99.02.R19399.img	Original firmware image to be signed
pkg_A30R_V01.99.02.R19399_0_0_TMS.sig	The generated signature file is used for AppUpgradeTool upgrade and TMS upgrade

3.3.4 Precautions

1. The signed file must correspond one-to-one with the img file and cannot be mixed up;
2. After each firmware update, re-signing is required;
3. KeyPair.ini must match the public key already injected into the device, otherwise the upgrade will fail;
4. Do not close the tool during the signing process.